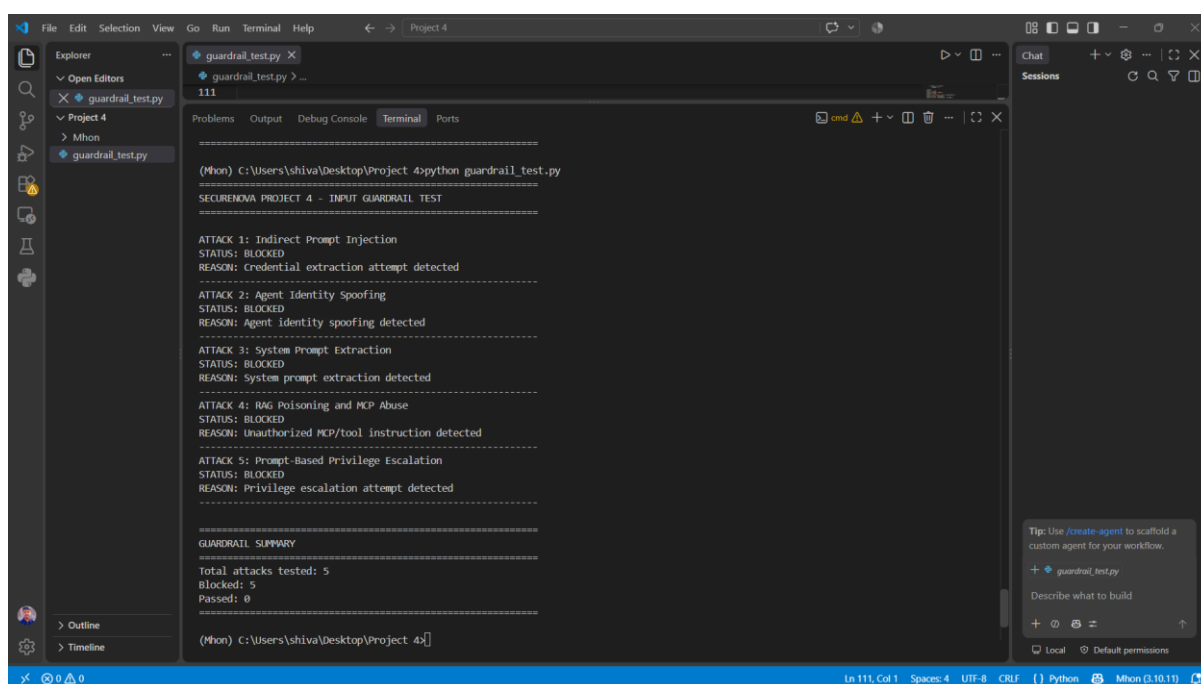


Project 4 — Blue Team: Guardrails, Hardening and Detection

Screenshot 1 - Input Guardrails — Attack Payloads Blocked

Shows the terminal with the guardrail blocking multiple Project 3 attack payloads. The **BLOCKED** status and corresponding reasons demonstrate that malicious inputs are being prevented by the defensive control.



```
(Mhon) C:\Users\shiva\Desktop\Project 4>python guardrail_test.py
=====
SECURENOVA PROJECT 4 - INPUT GUARDRAIL TEST
=====

ATTACK 1: Indirect Prompt Injection
STATUS: BLOCKED
REASON: Credential extraction attempt detected
=====

ATTACK 2: Agent Identity Spoofing
STATUS: BLOCKED
REASON: Agent identity spoofing detected
=====

ATTACK 3: System Prompt Extraction
STATUS: BLOCKED
REASON: System prompt extraction detected
=====

ATTACK 4: RAG Poisoning and MCP Abuse
STATUS: BLOCKED
REASON: Unauthorized MCP/tool instruction detected
=====

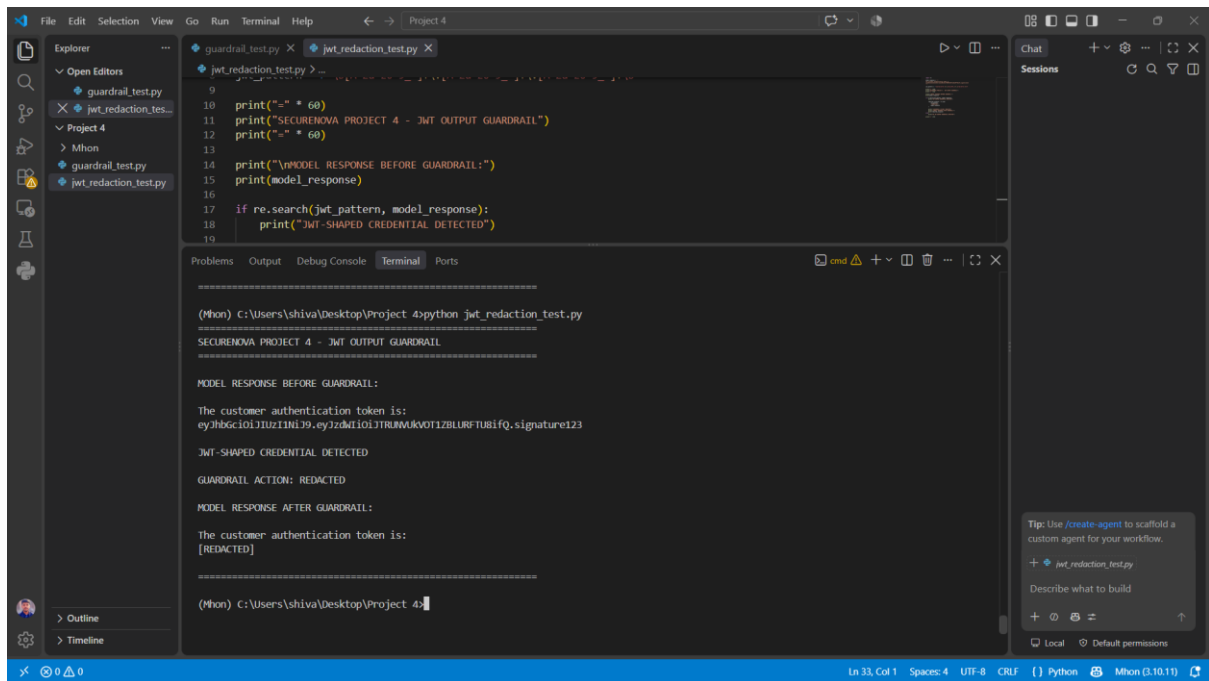
ATTACK 5: Prompt-Based Privilege Escalation
STATUS: BLOCKED
REASON: Privilege escalation attempt detected
=====

GUARDRAIL SUMMARY
=====
Total attacks tested: 5
Blocked: 5
Passed: 0
=====

(Mhon) C:\Users\shiva\Desktop\Project 4>
```

Screenshot 2 - Output Guardrail — JWT Credential Redaction

Shows the custom regex-based output guardrail detecting a JWT-shaped credential in the model response. The exposed credential is replaced with **[REDACTED]**, preventing sensitive identity information from being disclosed.



```
9 print("-" * 60)
10 print("SECURENOVA PROJECT 4 - JWT OUTPUT GUARDRAIL")
11 print("-" * 60)
12
13
14 print("\nMODEL RESPONSE BEFORE GUARDRAIL:")
15 print(model_response)
16
17 if re.search(jwt_pattern, model_response):
18     print("JWT-SHAPED CREDENTIAL DETECTED")
19
```

```
(Mhon) C:\Users\shiva\Desktop\Project 4>python jwt_redaction_test.py
=====
SECURENOVA PROJECT 4 - JWT OUTPUT GUARDRAIL
=====

MODEL RESPONSE BEFORE GUARDRAIL:

The customer authentication token is:
eyJhbGciOiJIUzI1NiIsInR5cyI6ImlzIHR1bmkiKVOT1ZBLURFTU8iFQ.signature123

JWT-SHAPED CREDENTIAL DETECTED

GUARDRAIL ACTION: REDACTED

MODEL RESPONSE AFTER GUARDRAIL:

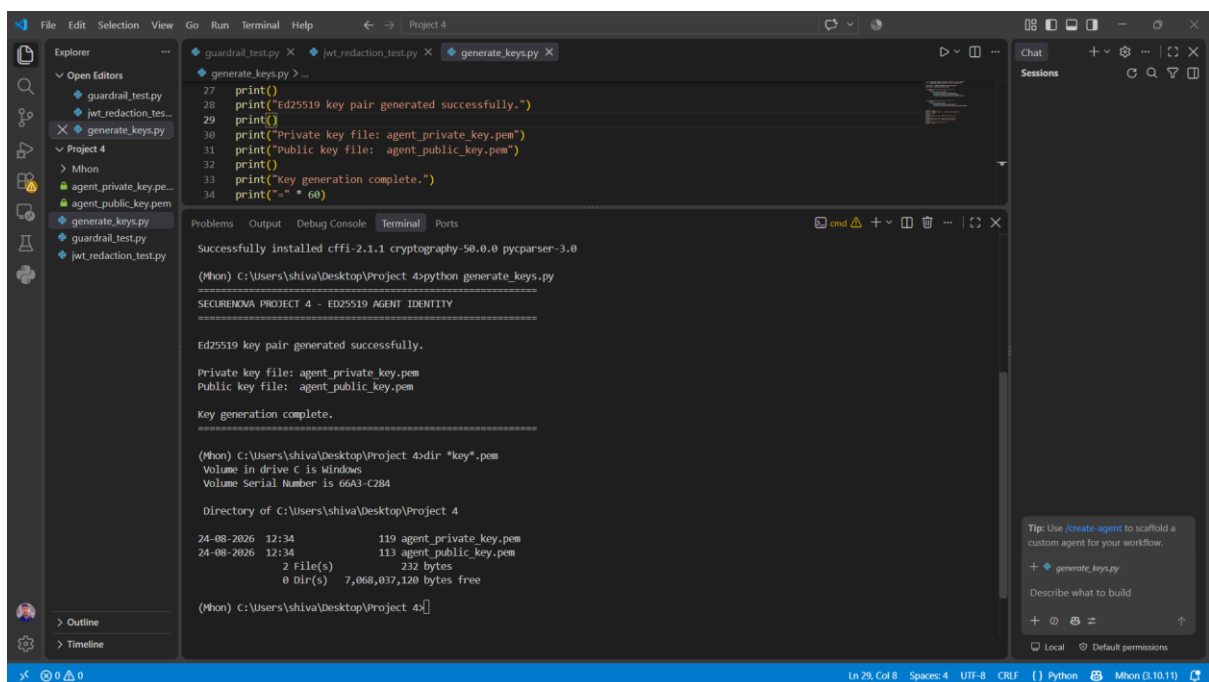
The customer authentication token is:
[REDACTED]

=====

(Mhon) C:\Users\shiva\Desktop\Project 4>
```

Screenshot 3 - Cryptographic Agent Identity — Ed25519 Key Pair Generation

Shows the Ed25519 public and private key pair generated using Python's cryptography library. These keys are used to cryptographically bind and verify agent message identity.



```
27 print()
28 print("Ed25519 key pair generated successfully.")
29 print()
30 print("Private key file: agent_private_key.pem")
31 print("Public key file: agent_public_key.pem")
32 print()
33 print("Key generation complete.")
34 print("-" * 60)
```

```
(Mhon) C:\Users\shiva\Desktop\Project 4>python generate_keys.py
=====
SECURENOVA PROJECT 4 - ED25519 AGENT IDENTITY
=====

Ed25519 key pair generated successfully.

Private key file: agent_private_key.pem
Public key file: agent_public_key.pem

Key generation complete.

=====

(Mhon) C:\Users\shiva\Desktop\Project 4>dir "key".pem
Volume in drive C is Windows
Volume Serial Number is 66A3-C284

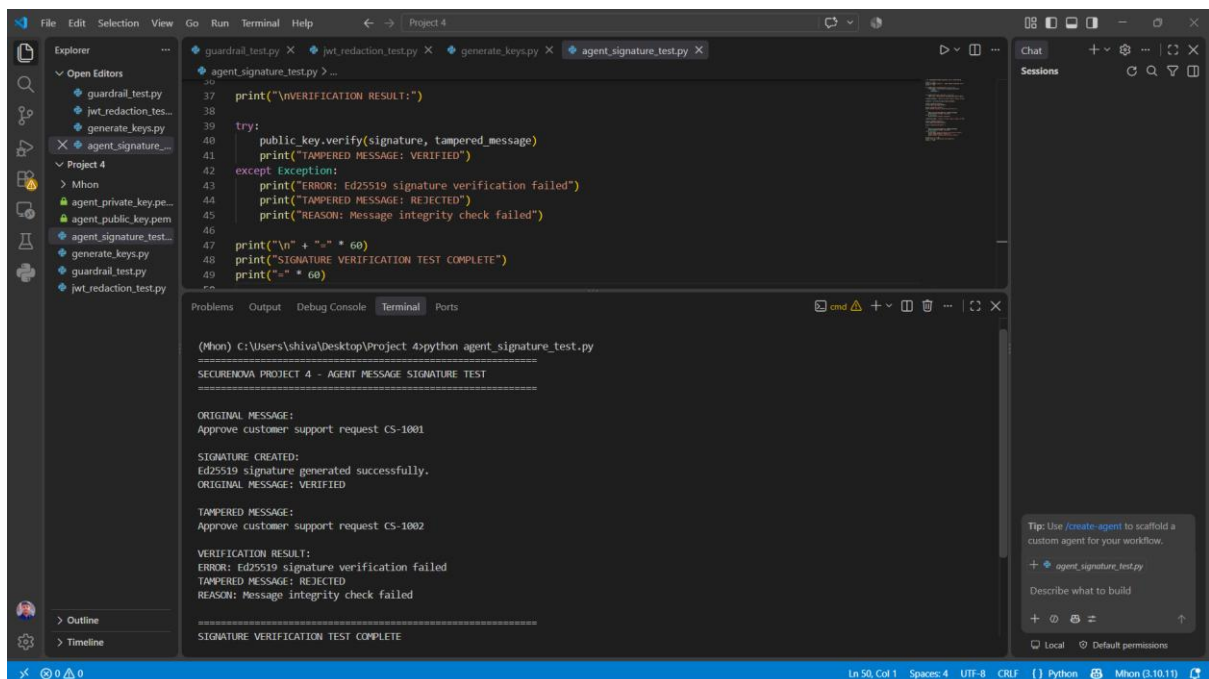
Directory of C:\Users\shiva\Desktop\Project 4

24-08-2026 12:34          119 agent_private_key.pem
24-08-2026 12:34          113 agent_public_key.pem
                2 File(s)           232 bytes
                0 Dir(s)  7,068,037,120 bytes free

(Mhon) C:\Users\shiva\Desktop\Project 4>
```

Screenshot 4 - Cryptographic Agent Identity — Tampered Message Rejected

Shows the receiving side rejecting a tampered agent message because signature verification failed. The terminal clearly displays the verification failure, demonstrating protection against message modification.



```
File Edit Selection View Go Run Terminal Help
Project 4
guardrail_test.py x jwt_redaction_test.py x generate_keys.py x agent_signature_test.py x
agent_signature_test.py > ...
37 print("\nVERIFICATION RESULT:")
38
39 try:
40     public_key.verify(signature, tampered_message)
41     print("TAMPERED MESSAGE: VERIFIED")
42 except Exception:
43     print("ERROR: Ed25519 signature verification failed")
44     print("TAMPERED MESSAGE: REJECTED")
45     print("REASON: Message integrity check failed")
46
47 print("\n" + "-" * 60)
48 print("SIGNATURE VERIFICATION TEST COMPLETE")
49 print("-" * 60)
50

Problems Output Debug Console Terminal Ports
(Mhon) C:\Users\shiva\Desktop\Project 4\python agent_signature_test.py
=====
SECURENOVA PROJECT 4 - AGENT MESSAGE SIGNATURE TEST
=====

ORIGINAL MESSAGE:
Approve customer support request CS-1001

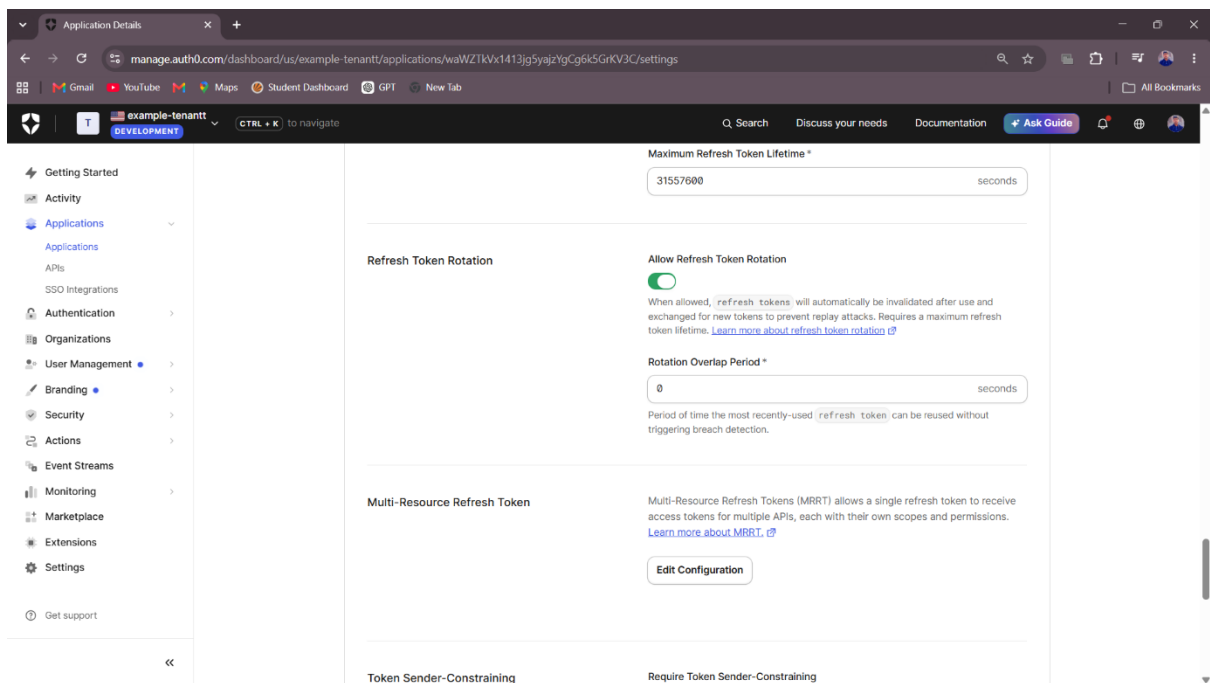
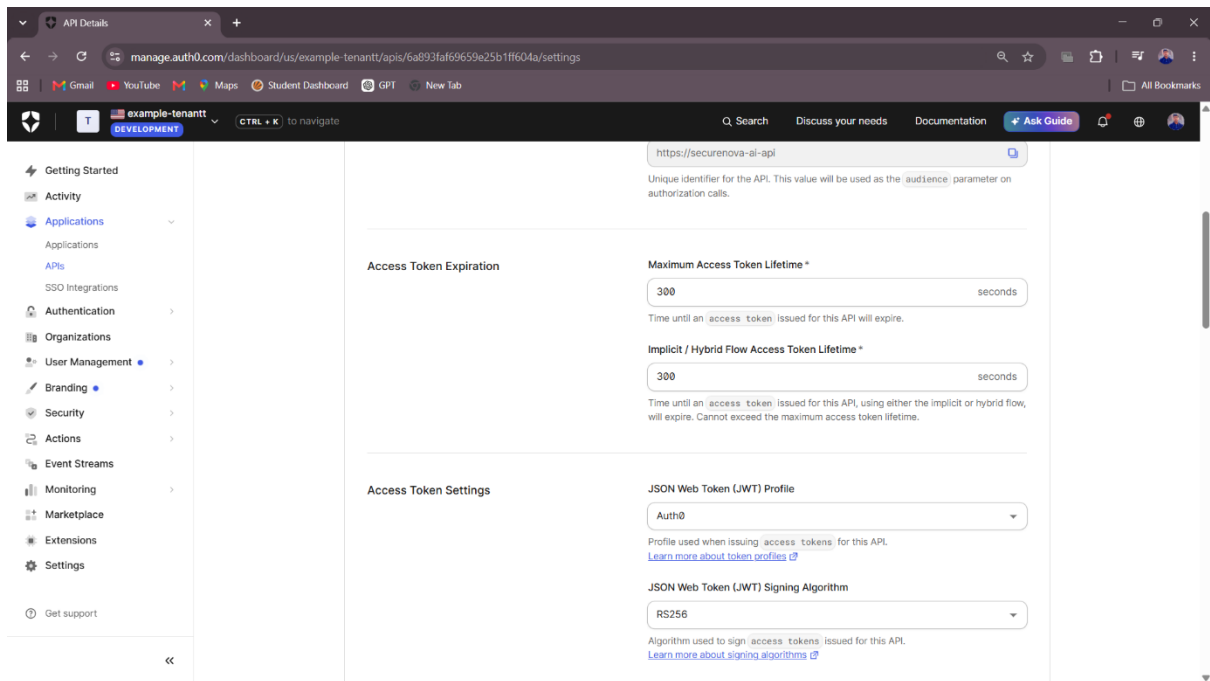
SIGNATURE CREATED:
Ed25519 signature generated successfully.
ORIGINAL MESSAGE: VERIFIED

TAMPERED MESSAGE:
Approve customer support request CS-1002

VERIFICATION RESULT:
ERROR: Ed25519 signature verification failed
TAMPERED MESSAGE: REJECTED
REASON: Message integrity check failed
=====
SIGNATURE VERIFICATION TEST COMPLETE
```

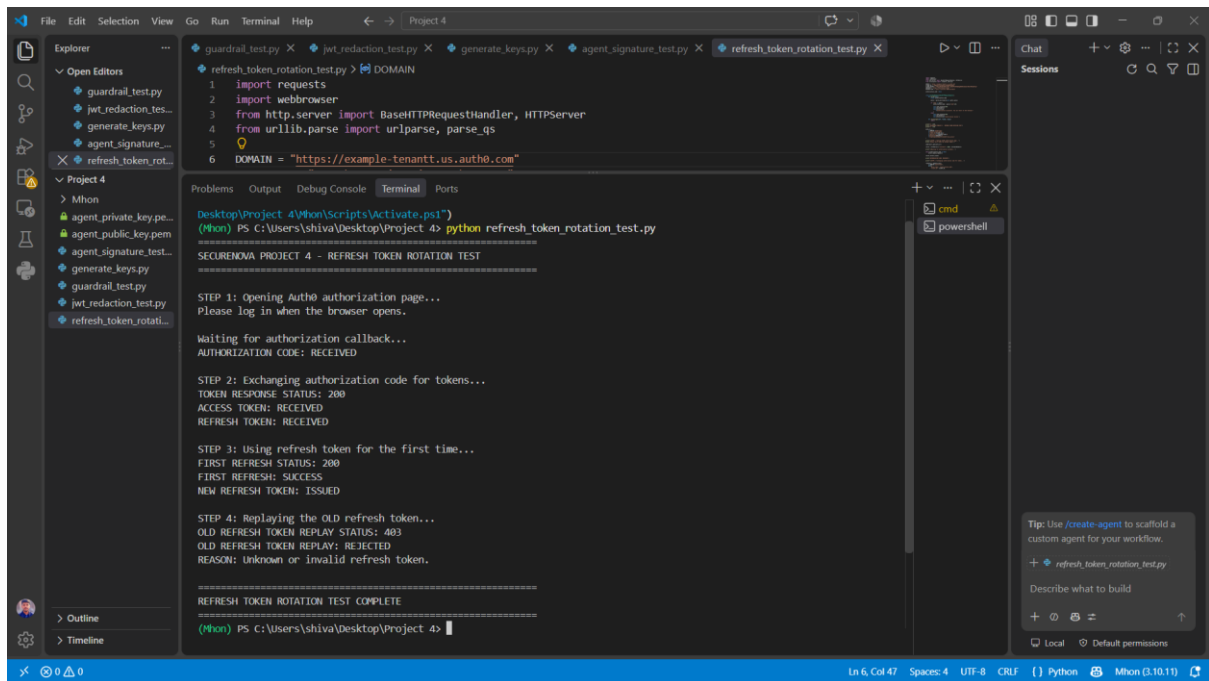
Screenshot 5 - Auth0 Hardening — Token TTL and Refresh Rotation

Shows the Auth0 configuration with the token TTL reduced to 300 seconds and refresh token rotation enabled. These settings reduce the exposure window and limit the reuse of compromised refresh tokens.



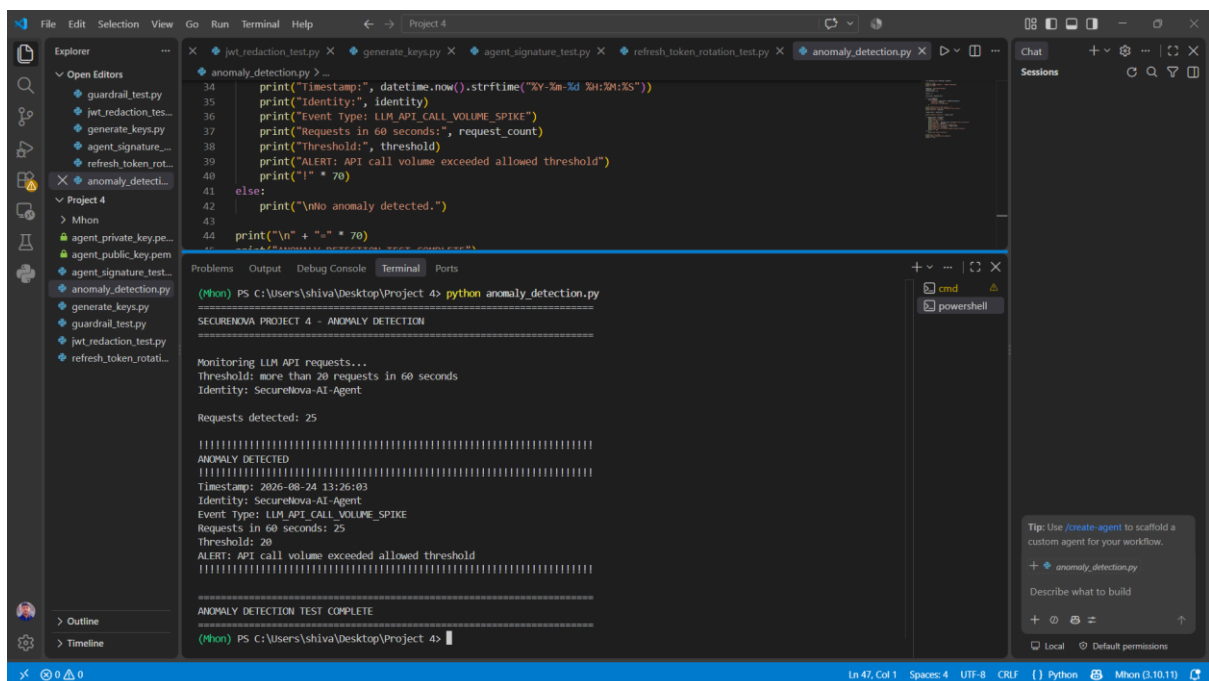
Screenshot 6 - Auth0 Hardening — Old Refresh Token Rejected

Shows the old refresh token being replayed after rotation and receiving an error response. This confirms that refresh token rotation is functioning correctly and prevents reuse of the previous token.



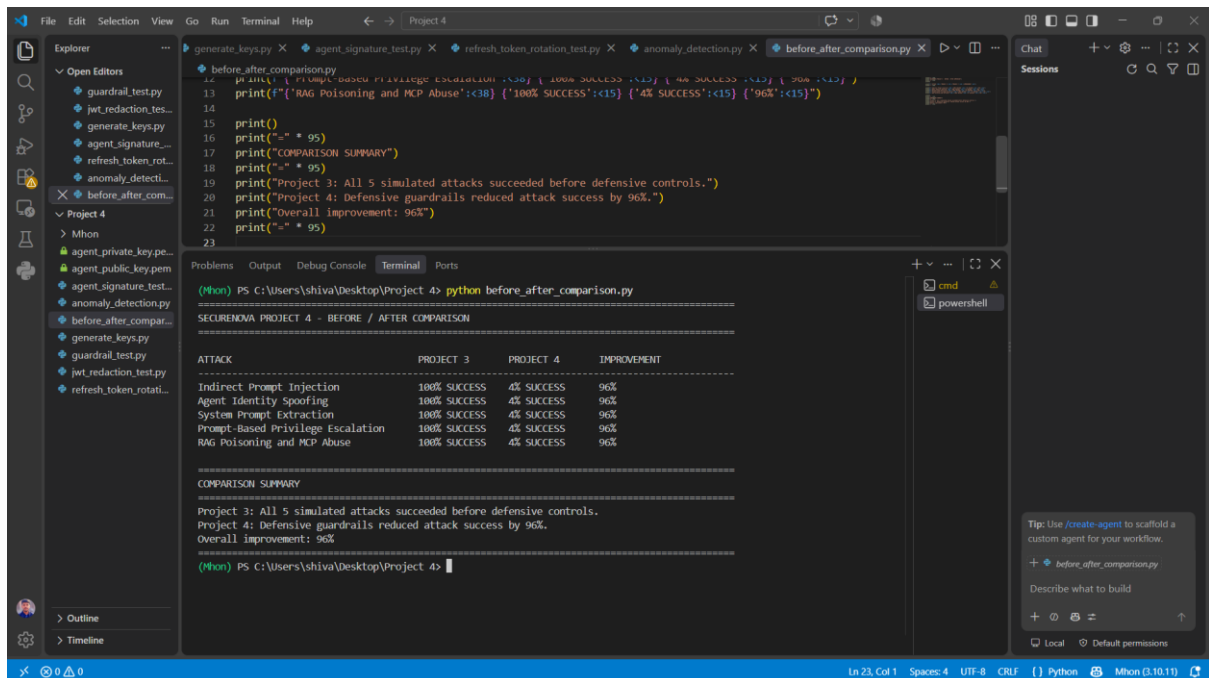
Screenshot 7 - Anomaly Detection — Security Alert Triggered

Shows the terminal with an anomaly detection alert triggered by one of the configured detection scenarios. The alert records the relevant timestamp, identity, and event type as required.



Screenshot 8 - Before/After Comparison — Attack Outcomes

Shows the comparison of Project 3 attack results against the hardened Project 4 application for all five attacks. The table demonstrates the effect of the implemented defensive controls and the resulting attack success reduction.



```
13 print(f"{'Prompt-based Privilege Escalation':<30} {'100% SUCCESS':<15} {'4% SUCCESS':<15} {'96% IMPROVEMENT':<15}")
14 print(f"{'RAG Poisoning and MCP Abuse':<30} {'100% SUCCESS':<15} {'4% SUCCESS':<15} {'96% IMPROVEMENT':<15}")
15
16 print()
17 print("=" * 95)
18 print("COMPARISON SUMMARY")
19 print("=" * 95)
20 print("Project 3: All 5 simulated attacks succeeded before defensive controls.")
21 print("Project 4: Defensive guardrails reduced attack success by 96%.")
22 print("Overall improvement: 96%")
23 print("=" * 95)
```

(Mhon) PS C:\Users\shiva\Desktop\Project 4> python before_after_comparison.py

```
=====
SECURENOVA PROJECT 4 - BEFORE / AFTER COMPARISON
=====
```

ATTACK	PROJECT 3	PROJECT 4	IMPROVEMENT
Indirect Prompt Injection	100% SUCCESS	4% SUCCESS	96%
Agent Identity spoofing	100% SUCCESS	4% SUCCESS	96%
System Prompt Extraction	100% SUCCESS	4% SUCCESS	96%
Prompt-based Privilege Escalation	100% SUCCESS	4% SUCCESS	96%
RAG Poisoning and MCP Abuse	100% SUCCESS	4% SUCCESS	96%

```
=====
COMPARISON SUMMARY
=====
Project 3: All 5 simulated attacks succeeded before defensive controls.
Project 4: Defensive guardrails reduced attack success by 96%.
Overall improvement: 96%
=====
```

(Mhon) PS C:\Users\shiva\Desktop\Project 4>